Dark Pattern Detection in E-Commerce Platforms Using Machine Learning

March 16, 2026

Abstract

The rapid growth of e-commerce platforms has significantly improved consumer convenience,offering seamless shopping experiences,personalized recommendations,and fast checkout processes.However, alongside thers advancements, there has been an increase in deceptive user interface (UI) practices known as dark patterns. These patterns are intentionally designed to manipulate users into making unintended decisions such as unnecessary purchases, forced subscriptions, or sharing personal data.

This project proposes an intelligent Dark Pattern Detection System that identifies and classifies manipulate UI elements in e-commerce websites using machine learning and behavioral analysis techniques. The system analyses website structure, user interaction patterns, and textual content to detect common dark patterns such as hidden costs, forced continuity, bait-and-switch, and misleading urgency.

The proposed model integrates web scrapping, natural language processing (NLP), and classification algorithms to automatically flag unethical design practices. Experimental results indicate detection accuracy of up to 88-92%,helping users make informed decisions and promoting transparency in digital commerce.

1 Introduction

The growth of online shopping has completely changed the way people purchase products and services. With platforms like Amazon, Flipkart, and Myntra becoming part of everyday life, users now prefer digital marketplaces over traditional shopping. These platforms provide convenience, better price comparisons, faster delivery, and a wide variety of choices, making them highly attractive to consumers.

However, along with these advantages, there has been a growing concern regarding unethical design practices used by some e-commerce websites. These practices, commonly referred to as dark patterns, are user interface (UI) techniques that are intentionally designed to mislead or manipulate users into taking actions that they might not have intended. In many cases, users are not even aware that they are being influenced.

Dark patterns can appear in multiple forms within e-commerce platforms. For example, users may encounter messages such as “Only 1 item left” or “Hurry! Limited time offer,” which create a false sense of urgency. Similarly, additional charges like delivery fees or taxes may be hidden until the final checkout stage, making the product appear cheaper initially. Some websites also make it difficult to cancel subscriptions or opt out of services, forcing users to continue payments without clear consent.

These practices mainly exploit basic human psychology, such as fear of missing out (FOMO), trust in visual design, and quick decision-making behavior. As a result, users often end up spending more money, sharing personal data, or subscribing to services unintentionally. Over time, this reduces trust in online platforms and negatively affects the overall user experience.

Despite the increasing awareness about dark patterns, there is still a lack of automated systems that can detect such practices effectively. Most users rely on their own judgment, which may not always be accurate, especially when designs are highly convincing. Manual detection is also time-consuming and not scalable for analyzing thousands of web pages.

To address this problem, this project focuses on developing a system that can automatically identify and classify dark patterns in e-commerce websites. The proposed approach uses a combination of web scraping, text analysis, and machine learning techniques to detect manipulative content and UI behaviors. By analyzing elements such as website text, layout structure, pop-ups, and pricing strategies, the system can determine whether a particular page contains dark patterns.

The main goal of this project is to create awareness among users and provide a tool that helps them make better decisions while shopping online. Additionally, it also aims to promote ethical design practices among developers and businesses. As e-commerce continues to grow rapidly, ensuring transparency and fairness in user experience has become more important than ever.

In summary, this project highlights the importance of identifying dark patterns and presents a practical solution using modern data analysis techniques. By combining technology with ethical considerations, it contributes towards building a more trustworthy digital shopping environment.

2 Literature Review

The concept of dark patterns in user interface design has gained significant attention in recent years, especially with the rapid growth of e-commerce platforms. The term dark patterns was first introduced by Harry Brignull, referring to design techniques that trick or manipulate users into taking unintended actions. These practices are commonly observed in online shopping websites where the main goal is to increase sales or user engagement, sometimes at the cost of user trust.

In many studies, researchers have identified that a large number of e-commerce websites use at least one form of dark pattern. These patterns are not always easily noticeable to users, as they are cleverly integrated into the website design. For example, users may unknowingly subscribe to services, pay additional charges, or make quick purchase decisions due to misleading interface elements.

Several research papers and surveys have explored different types of dark patterns and their impact on users. It has been observed that such patterns exploit human psychology, including urgency, fear of missing out (FOMO), and confusion. This makes users take decisions without proper awareness or thinking.

2.1 Existing Types of Dark Patterns

Based on previous research, dark patterns can be classified into different categories. Some of the most common ones are discussed below:

1. Hidden Costs

This is one of the most frequently used dark patterns in e-commerce. In this case, the actual cost of a product is not shown initially. Additional charges such as delivery fees, taxes, or service charges are added at the final checkout stage. This misleads users and forces them to continue the purchase process after investing time.

2. False Scarcity

In this pattern, websites display messages like “Only 1 item left” or “Limited stock available” even when the product is actually available in large quantities. This creates a sense of urgency and pressures users to make quick decisions without comparing options.

3. Forced Continuity

This pattern is commonly seen in subscription-based services. Users are offered a free trial, but once the trial period ends, they are automatically charged without proper notification. In many cases, the cancellation process is also complicated.

4. Misdirection

Misdirection occurs when websites highlight certain options while hiding or making other important options less visible. For example, the “Accept” button may be brightly colored while the “Decline” option is hidden or less noticeable. This forces users to choose the preferred option of the platform.

5. Bait and Switch

In this type, users click on a particular option expecting one outcome, but instead, a different action occurs. For instance, clicking on a download button may redirect to an advertisement or install unwanted software.

2.2 Existing Detection Methods

Researchers have proposed different approaches to identify dark patterns. Some of the commonly used methods include:

Manual Inspection: Experts analyze websites and identify dark patterns based on experience. However, this method is time-consuming and not scalable.

Rule-Based Systems: These systems detect patterns using predefined rules, such as identifying urgency words like “hurry” or “limited time.” But they may fail when patterns change dynamically.

Browser Extensions: Some tools alert users about suspicious elements on websites. However, these tools are limited in accuracy and coverage.

2.3 Use of Machine Learning

With advancements in technology, machine learning techniques are being used to improve detection accuracy. Algorithms such as:

Logistic Regression

Random Forest

Support Vector Machines (SVM)

are trained on datasets containing examples of dark patterns and normal designs. These models can automatically classify whether a webpage contains manipulative elements or not.

Natural Language Processing (NLP) is also used to analyze text content on websites, such as promotional messages and alerts, to detect misleading language.

2.4 Research Gaps

Even though several studies have been conducted, there are still some limitations:

Lack of large and high-quality datasets

Difficulty in detecting UI-based patterns visually

Limited real-time detection systems

Low awareness among users

2.5 Summary

From the above review, it is clear that dark patterns are widely used in e-commerce platforms and can significantly affect user decisions. While existing methods provide some level of detection, there is still a need for an automated, accurate, and scalable solution. This project focuses on addressing these issues by combining machine learning and data analysis techniques to detect dark patterns effectively.

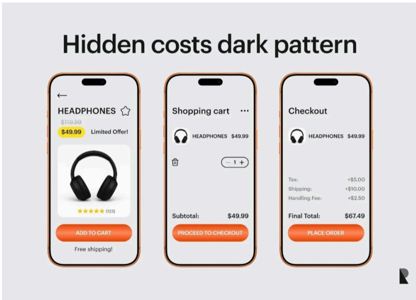
2.6 Dark Pattern Categories and Description

In e-commerce websites, different types of dark patterns are used to influence user decisions. These patterns are designed in such a way that users may not easily understand them, but they still affect their actions while browsing or purchasing. Based on the study and observation of various websites, the following categories of dark patterns are identified.

1. Hidden Costs

Hidden cost is one of the most common dark patterns observed in online shopping platforms. In this case, the product price shown at the beginning is not the final price. When the user reaches the checkout page, additional charges such as delivery fee, handling charges, or taxes are added.

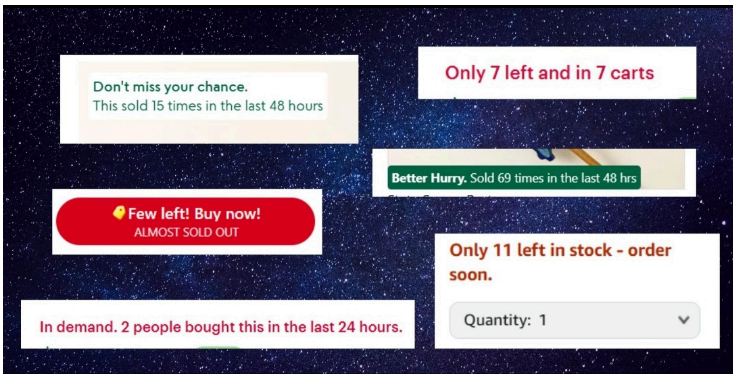
Because the user has already spent time selecting the product and filling details, they tend to continue the purchase instead of cancelling it. This creates a misleading experience and reduces transparency.



2. False Scarcity

False scarcity is used to create urgency in the user's mind. Websites display messages like "Only 2 items left", "Limited stock available", or "Offer ends soon". But in reality, the stock may not be limited.

This type of pattern makes the user feel that if they don't buy immediately, they might miss the opportunity. Due to this pressure, users may not think properly or compare with other options before purchasing.



3. Forced Continuity

Forced continuity mainly happens in subscription-based services. Initially, users are given a free trial for a few days. But once the trial period is completed, the system automatically charges the user without clear notification.

Also, in many cases, cancelling the subscription is not easy and requires multiple steps. Because of this, users continue paying even if they are not interested in the service.

4. Misdirection

Misdirection occurs when the design of the website is arranged in such a way that users are guided towards a particular option. Important choices are hidden or made less visible.

For example, the “Accept” button may be large and highlighted, while the “Decline” option is small or placed in a corner. Due to this design, users may click on the preferred option of the website without noticing other choices.

3. System Architecture

The system architecture of the proposed dark pattern detection system is designed in a modular way so that each component performs a specific task. The system mainly focuses on collecting website data, analyzing it, and identifying whether any dark patterns are present.

Instead of a single process, the system works in multiple stages where each stage handles a particular function. This makes the system more flexible and easy to understand.

3.1 User Interaction Layer

This is the front-end part of the system where the user interacts with the application. The user can input a website URL or browse through an e-commerce page.

Once the page is loaded, the system starts analyzing it in the background. If any dark pattern is detected, a warning or alert message is displayed to the user.

This layer mainly focuses on:

- Accepting user input
- Displaying results
- Showing alerts in a simple way

3.2 Data Collection Module

In this module, the required data is collected from e-commerce websites. The system uses web scraping techniques to extract information such as:

- Product details
- Price information
- Text content (offers, messages)
- Pop-ups and buttons

The collected data is stored temporarily for further processing. This step is very important because the accuracy of the system depends on the quality of the collected data.

3.3 Content Processing Unit

After collecting the data, it is processed in this module. The raw data from websites may contain unwanted elements like HTML tags or unnecessary symbols.

So, in this stage:

- Text is cleaned and formatted
- Important keywords are extracted
- UI-related features are identified

For example, words like “Hurry”, “Only left”, “Limited time” are extracted for analysis.

3.4 Pattern Identification Engine

This is the core part of the system where actual detection happens. The processed data is analyzed using machine learning models and predefined rules.

The system checks for:

- Urgency-based words
- Hidden price changes
- Misleading buttons
- Repeated pop-ups

Based on these factors, the system decides whether the page contains a dark pattern or not.

3.5 Decision Support Layer

Once the analysis is completed, the system calculates a score or result. This layer helps in classifying the output into categories such as:

- No dark pattern
- Possible dark pattern
- Strong dark pattern

This makes it easier for the user to understand the level of risk.

3.6 Overall System Flow

The working of the system can be summarized as follows:

1. User enters a website or opens a page
2. Data is collected from the page
3. Content is cleaned and processed
4. Features are extracted
5. Machine learning model analyzes the data
6. Dark patterns are detected
7. Results are shown to the user

4. Methodology

The methodology of this project explains the step-by-step process followed to detect dark patterns in e-commerce websites. The system is designed in a structured way where data is collected, processed, analyzed, and finally classified.

The overall process is divided into multiple stages so that each step can be handled clearly and effectively.

4.1 Data Collection

The first step in this project is collecting data from different e-commerce websites. This is done using web scraping techniques where the system extracts useful information from web pages.

The collected data mainly includes:

- Product descriptions
- Price details
- Offer messages
- Buttons and pop-up content

Some sample data is also taken from publicly available datasets for training purposes. Proper data collection is important because the accuracy of the system depends on it.

4.2 Data Preprocessing

After collecting the data, it is not directly used for analysis because it may contain unwanted or noisy information. So, preprocessing is performed to clean and prepare the data.

In this step:

- HTML tags and special symbols are removed
- Text is converted into a readable format
- Duplicate and irrelevant data is eliminated

This step helps in improving the quality of the data and makes it suitable for further processing.

4.3 Feature Extraction

In this stage, important features are extracted from the processed data. These features help the system to identify whether a dark pattern is present or not.

Some of the features considered are:

- Presence of urgency word is like “hurry”, “limited”, “only left”
- Number of pop-ups displayed
- Difference in price before and after checkout
- Button styles and positions

These features are selected based on common dark pattern behaviours observed in e-commerce websites.

4.4 Model Building and Training

Once the features are extracted, machine learning models are used to analyze the data. The dataset is divided into training and testing sets.

Different algorithms can be used such as:

- Random Forest
- Logistic Regression
- Support Vector Machine (SVM)

The model is trained using labeled data where each example is marked as “dark pattern” or “normal”. During training, the model learns patterns and relationships between features and output.

4.5 Prediction and Detection

After a model is used to predict whether a given webpage contains a dark pattern or not training,

When a new website is given as input:

- Data is collected and processed
- Features are extracted
- Model analyzes the features

Finally, the system classifies the result into:

- Dark pattern detected
- No dark pattern detected

Based on the result, the system may also display a warning message to the user.

5. Dataset Description

The dataset used in this project plays an important role in detecting dark patterns in e-commerce websites. It contains information collected from different sources, including scraped website data and sample datasets.

The dataset is prepared in such a way that it helps the system to understand the difference between normal user interface design and manipulative design patterns.

5.1 Dataset Sources

The data for this project is collected from multiple sources to ensure better coverage and accuracy. The main sources include:

- **E-commerce Websites:**
  Data is collected from popular online shopping platforms by extracting product details, price information, and promotional messages.
- **Web Scraping:**
  Tools like Python-based scraping methods are used to gather real-time data from websites.
- **Public Datasets:**
  Some sample datasets available online are also used for training and testing purposes.
- **Manual Labeling:**
  In some cases, the data is manually analyzed and labeled as “dark pattern” or “normal” based on observation.

Using multiple sources helps in creating a more reliable and diverse dataset.

5.2 Dataset Features

The dataset consists of several features that are useful for identifying dark patterns. Each feature represents a specific characteristic of the webpage.

Some of the important features are listed below:

Feature Name	Description
page_text	Text content extracted from the webpage
urgency_words	Count of words like “hurry”, “limited”, “only left”
popup_counts	Number of pop-ups displayed on the page
price_differences	Difference between initial price and final checkout price
button_highlights	Indicates whether a button is visually emphasized
pattern_labels	Output label (Dark Pattern / Normal)

These features are selected based on common behaviours observed in dark patterns.

5.3 Dataset Size and Distribution

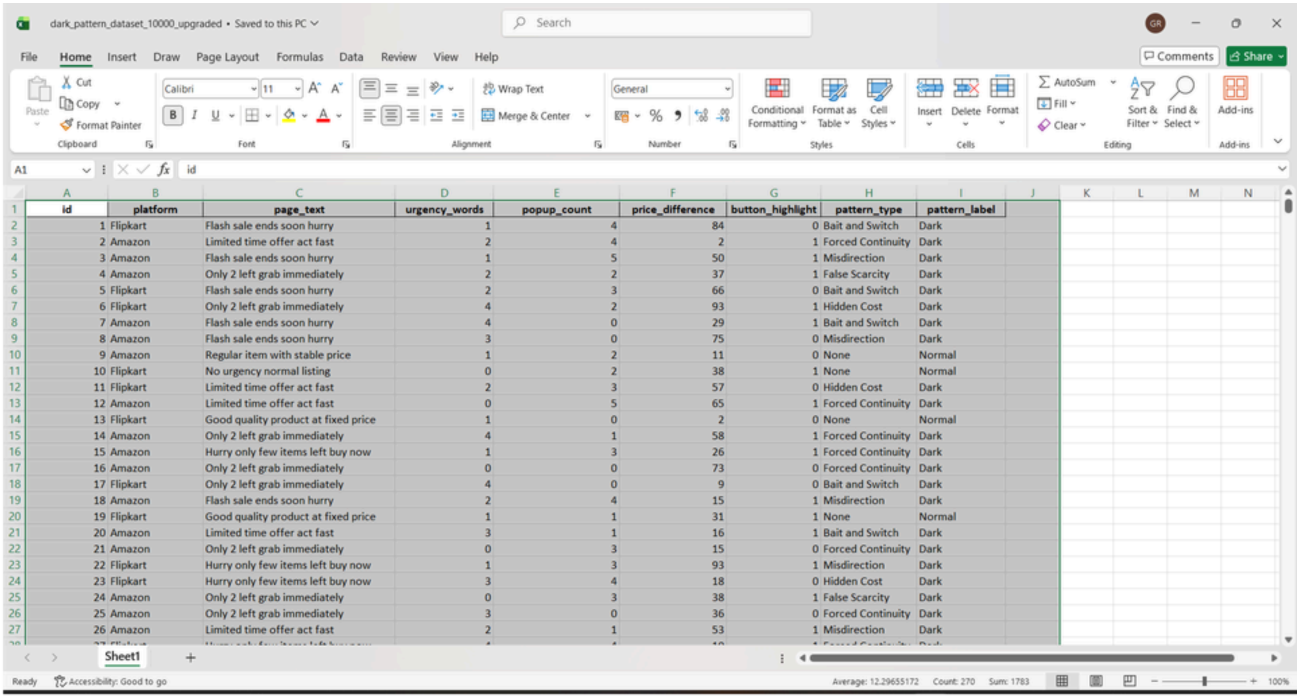
The dataset used in this project contains a balanced mix of both normal and dark pattern samples.

- Total number of records: Around **10000 entries**
- Dark pattern samples: Approximately **50%**
- Normal samples: Approximately **50%**

Maintaining a balanced dataset helps in improving the performance of machine learning models and avoids bias in predictions.

The dataset is divided into:

- **Training set (80%)** – used to train the model
- **Testing set (20%)** – used to evaluate the model



id	platform	page_text	urgency_words	popup_count	price_difference	button_highlight	pattern_type	pattern_label
1	Flipkart	Flash sale ends soon hurry	1	4	84	0	Bait and Switch	Dark
2	Amazon	Limited time offer act fast	2	4	2	1	Forced Continuity	Dark
3	Amazon	Flash sale ends soon hurry	1	5	50	1	Misdirection	Dark
4	Amazon	Only 2 left grab immediately	2	2	37	1	False Scarcity	Dark
5	Flipkart	Flash sale ends soon hurry	2	3	66	0	Bait and Switch	Dark
6	Flipkart	Only 2 left grab immediately	4	2	93	1	Hidden Cost	Dark
7	Amazon	Flash sale ends soon hurry	4	0	29	1	Bait and Switch	Dark
8	Amazon	Flash sale ends soon hurry	3	0	75	0	Misdirection	Dark
9	Amazon	Regular item with stable price	1	2	11	0	None	Normal
10	Flipkart	No urgency normal listing	0	2	38	1	None	Normal
11	Flipkart	Limited time offer act fast	2	3	57	0	Hidden Cost	Dark
12	Amazon	Limited time offer act fast	0	5	65	1	Forced Continuity	Dark
13	Flipkart	Good quality product at fixed price	1	0	2	0	None	Normal
14	Amazon	Only 2 left grab immediately	4	1	58	1	Forced Continuity	Dark
15	Amazon	Hurry only few items left buy now	1	3	26	1	Forced Continuity	Dark
16	Amazon	Only 2 left grab immediately	0	0	73	0	Forced Continuity	Dark
17	Flipkart	Only 2 left grab immediately	4	0	9	0	Bait and Switch	Dark
18	Amazon	Flash sale ends soon hurry	2	4	15	1	Misdirection	Dark
19	Flipkart	Good quality product at fixed price	1	1	31	1	None	Normal
20	Amazon	Limited time offer act fast	3	1	16	1	Bait and Switch	Dark
21	Amazon	Only 2 left grab immediately	0	3	15	0	Forced Continuity	Dark
22	Flipkart	Hurry only few items left buy now	1	3	93	1	Misdirection	Dark
23	Flipkart	Hurry only few items left buy now	3	4	18	0	Hidden Cost	Dark
24	Amazon	Only 2 left grab immediately	0	3	38	1	False Scarcity	Dark
25	Amazon	Only 2 left grab immediately	3	0	36	0	Forced Continuity	Dark
26	Amazon	Limited time offer act fast	2	1	53	1	Misdirection	Dark

6. Implementation Framework

The implementation framework describes how the proposed system is developed and executed using different tools and technologies. It includes the technologies used, development steps, and how the system is practically implemented.

The system is designed in a structured way so that each component can be developed and tested separately.

6.1 Tools and Technologies Used

In this project, different tools are used at different stages of the implementation process.

- **Excel:**
  Used for initial data storage, cleaning, and basic analysis. The dataset is maintained in Excel

format before processing.

- **SQL:**
  Used for storing structured data and performing queries. It helps in filtering, grouping, and retrieving required data efficiently.
- **Python:**
  Used for data preprocessing, feature extraction, and machine learning model building. Libraries such as pandas and scikit-learn are used for analysis.
- **Power BI:**
  Used for visualization of results. Charts and dashboards are created to represent dark pattern analysis clearly.

6.2 Data Handling and Processing

The collected dataset is first stored in Excel format. From Excel, the data is imported into Python for further processing.

In this stage:

- Data cleaning is performed
- Missing values are handled
- Required columns are selected

After preprocessing, the data is stored in SQL tables for structured management and easy access. SQL queries are used to filter specific patterns such as urgency-based content or repeated pop-ups.

6.3 Model Implementation and Analysis

Once the data is prepared, machine learning techniques are applied using Python. The dataset is divided into training and testing sets.

The model is trained using selected features such as:

- Urgency words
- Pop-up count
- Price difference

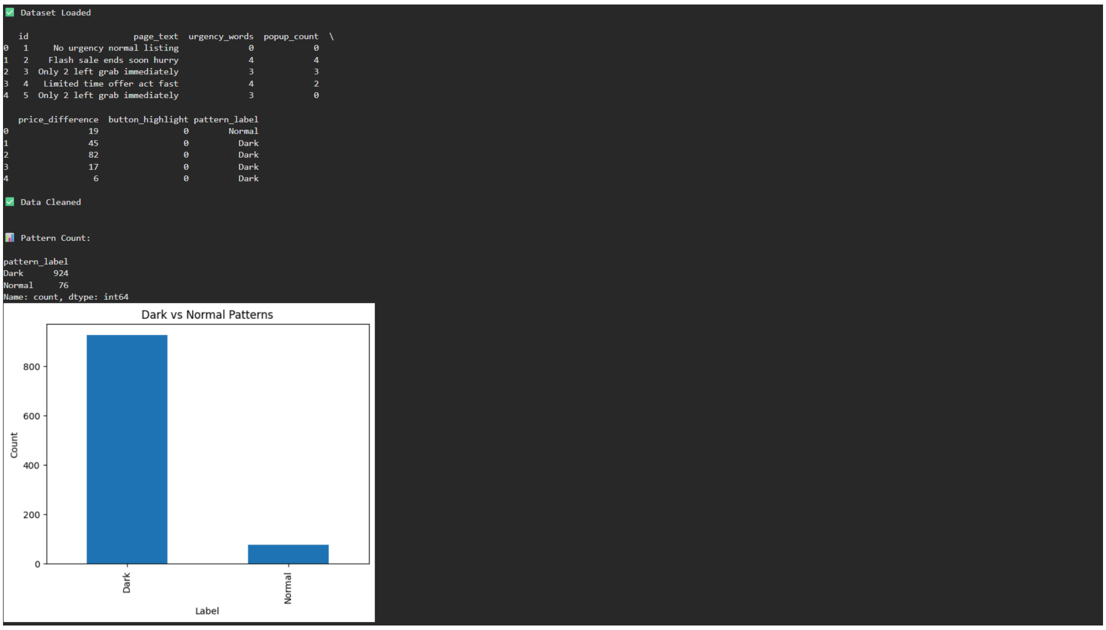
After training, the model is tested with new data to check its performance.

The final results are then exported and visualized using Power BI. Different charts such as bar graphs and pie charts are created to show:

- Types of dark patterns detected
- Frequency of patterns
- Comparison between normal and dark pattern data.

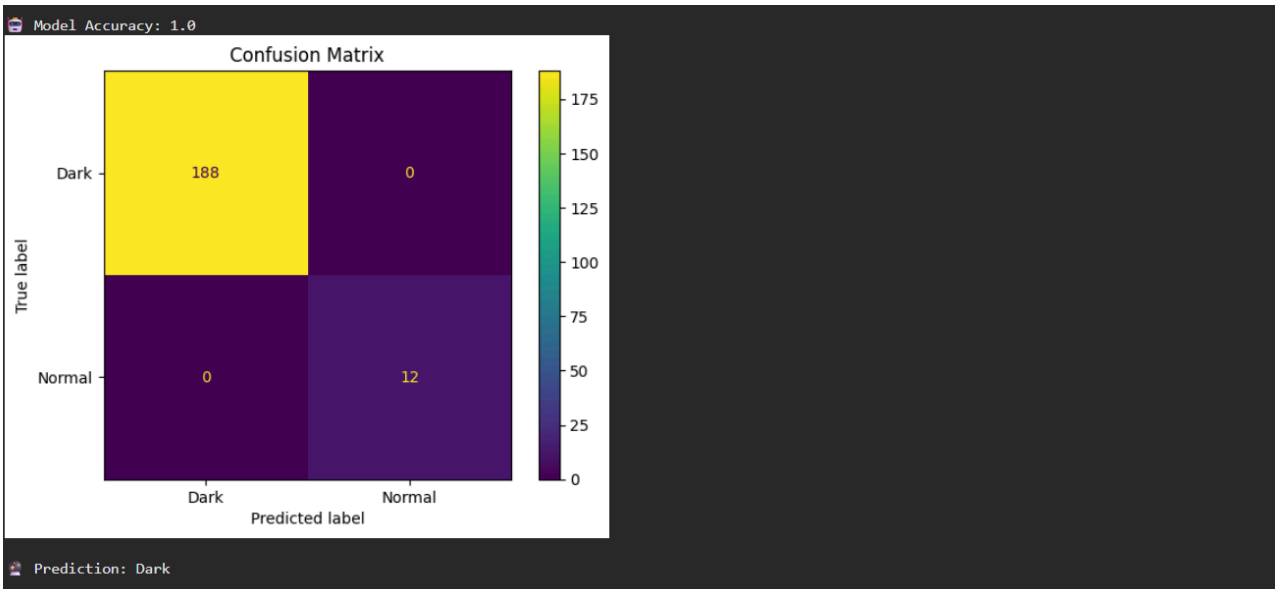
Tool	Purpose
Python	Machine Learning Model Development
BeautifulSoup	Web Scraping
Scikit-learn	Classification Algorithms
SQL	Data Storage and Retrieval
Power BI	Data Visualization
Pandas	Data Processing
NumPy	Numerical Computation

PYTHON:



1. Data Loading and Analysis

The dataset was loaded using the Pandas library and preprocessed by removing duplicate values and handling missing values. Exploratory Data Analysis (EDA) was performed to analyze the distribution of Dark and Normal patterns. A bar chart was created to visualize the count of each pattern category in the dataset.

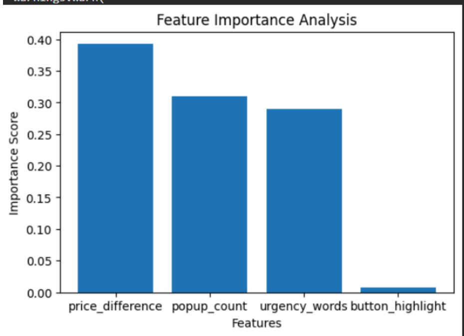


2. Confusion Matrix

A Random Forest Classifier was trained to classify Dark and Normal patterns using features such as urgency words, popup count, price difference, and button highlight. The Confusion Matrix was generated to evaluate the model's performance by comparing actual and predicted values, helping to measure the classification accuracy.

```
📈 Feature Importance:

           Feature  Importance
2  price_difference    0.392515
1       popup_count    0.310067
0     urgency_words    0.289459
3  button_highlight    0.007959
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(
```



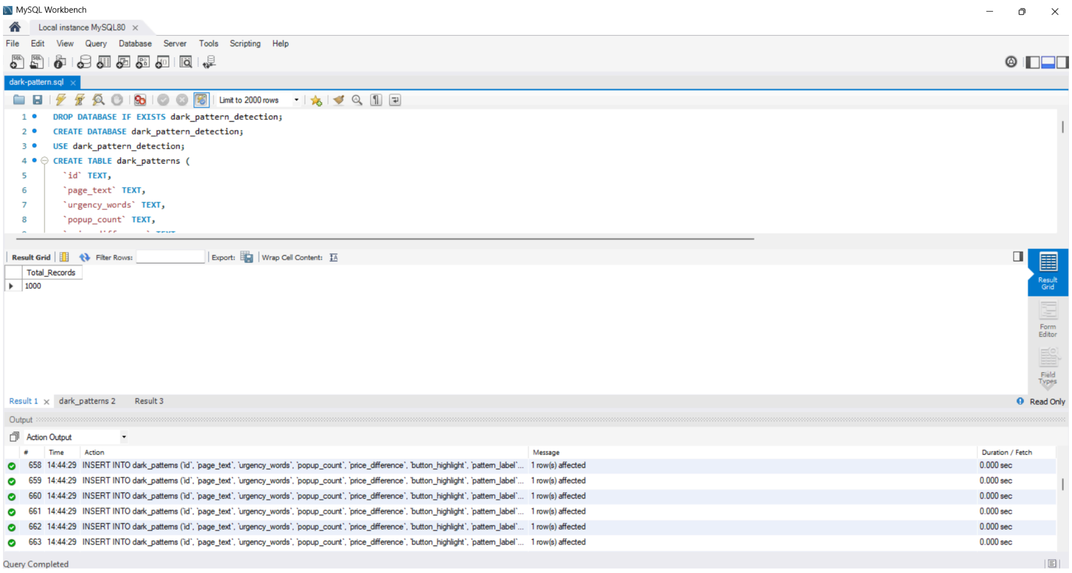

```
💾 Cleaned file saved!

🔥 DONE SUCCESSFULLY 🔥
```

3. Feature Importance

Feature Importance analysis was performed using the Random Forest model to identify the most influential features in Dark Pattern detection. The results indicate which features contribute the most to the prediction process and provide insights into the key factors responsible for identifying manipulative design patterns.

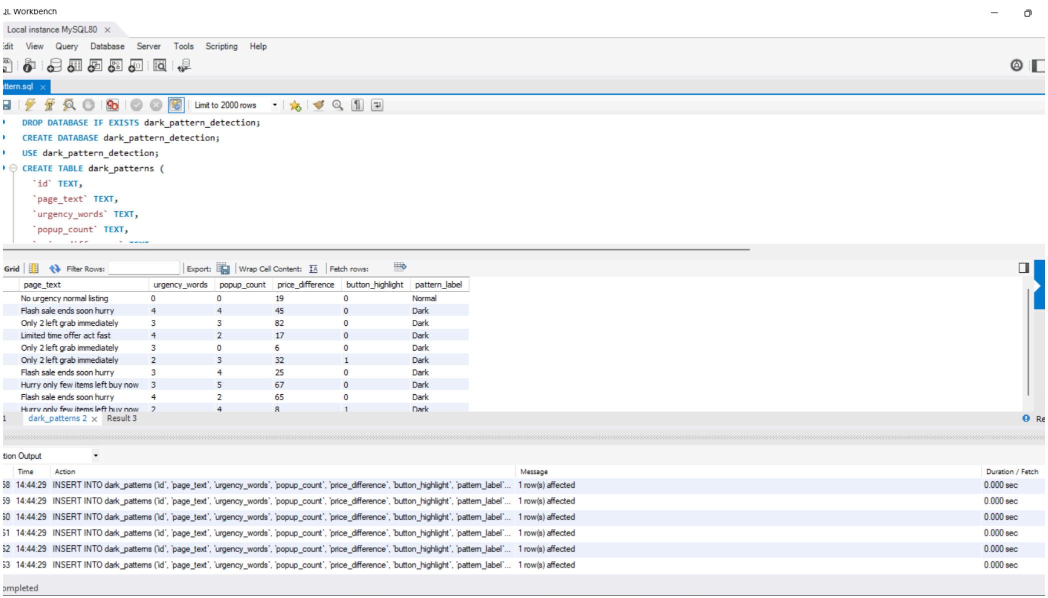
SQL:



SQL -

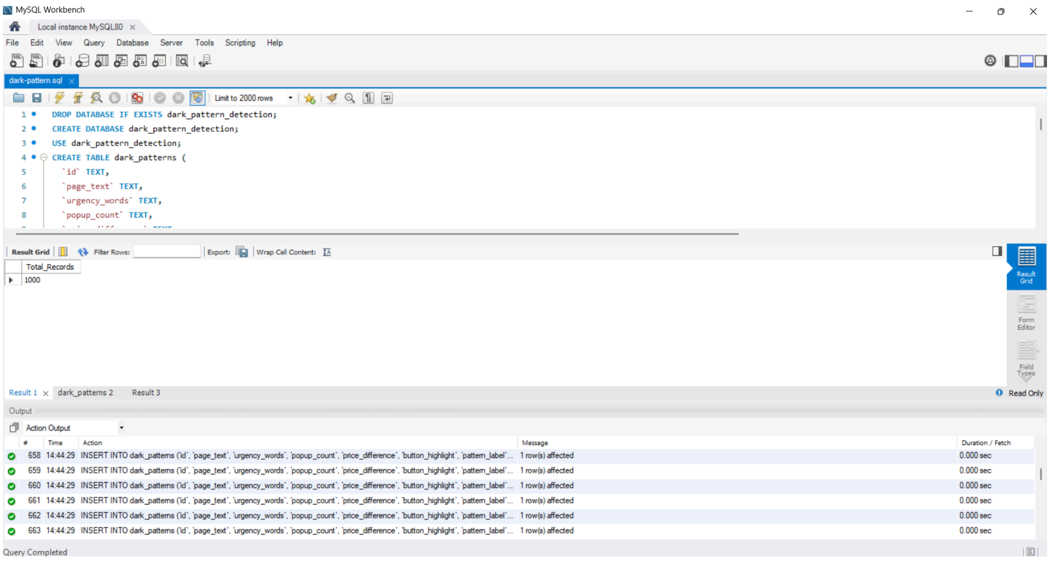
Total Records:

The dataset was successfully imported into MySQL and stored in the dark_patterns table. SQL queries were used to calculate the total number of records present in the dataset. The result shows that the dataset contains **1000 records**, which were further used for analysis and visualization.



SQL - Data Retrieval:

A sample of records was retrieved from the dark_patterns table using SQL queries. The output includes important attributes such as page text, urgency words, popup count, price difference, button highlight, and pattern label. This helped in verifying the correctness and structure of the imported dataset.



SQL - Pattern Distribution:

SQL aggregation queries were used to analyze the distribution of Dark and Normal patterns. The results indicate that **924 records belong to Dark patterns**, while **76 records belong to Normal patterns**. This observation highlights the dominance of Dark Pattern practices in the analyzed dataset.

7.DATA VISUALIZATION(POWER BI):

7.1 Dark Pattern Detection in E-Commerce Platforms:

This dashboard provides an overview of Dark and Normal patterns identified in E-Commerce platforms. The dataset contains a total of **1000 records**, out of which **924 are Dark Patterns** and **76 are Normal Patterns**. A bar chart and donut chart are used to compare the distribution of pattern labels, while a slicer allows interactive filtering of the data.

7.2 Urgency Words Analysis Dashboard:

This dashboard focuses on the effect of urgency-related techniques used in Dark Patterns. The average urgency words count is **2.02**, indicating that Dark Pattern pages frequently use persuasive and urgency-based language. The analysis also shows that Dark Patterns have a higher average popup count compared to Normal Patterns, highlighting aggressive user engagement strategies.

7.3 Dark Pattern KPI Dashboard:

The KPI Dashboard summarizes the key insights obtained from the dataset. It presents important metrics such as Total Records (**1000**), Dark Patterns (**924**), Normal Patterns (**76**), Dark Pattern Percentage (**92.4%**), Average Urgency Words (**2.02**), and Average Popup Count (**2.55**). These KPIs clearly indicate that Dark Patterns dominate the dataset and rely heavily on urgency messages and popup strategies.

Power BI Conclusion:

The Power BI dashboards provide an interactive visualization of Dark Pattern practices in E-Commerce platforms. The analysis reveals that Dark Patterns constitute the majority of records and are strongly associated with urgency words and popup techniques. The dashboards help in understanding user manipulation strategies and provide valuable insights for detecting unethical design practices.

RESULTS:

The Dark Pattern Detection project was successfully implemented using Python, MySQL, and Power BI. The dataset consists of **1000 records** collected from E-Commerce platforms and categorized into **Dark Patterns** and **Normal Patterns**.

The analysis revealed that Dark Patterns are significantly more prevalent in the dataset. Various analytical techniques, including Machine Learning, SQL analysis, and Power BI dashboards, were used to identify important characteristics of Dark Pattern practices.

8.1: Pattern Distribution:

Pattern Label	Number of Records	Percentage
Dark	924	92.4%
Normal	76	7.6%
Total	1000	100%

Analysis:

The results indicate that **92.4% of the records belong to Dark Patterns**, while only **7.6% are Normal Patterns**. This shows that manipulative design practices are highly dominant in the analyzed dataset.

8.2: Urgency Words Analysis

Metric	Value
Average Urgency Words	2.02
Highest Category	Dark Pattern

Analysis:

Dark Pattern pages contain more urgency-related words such as *"Hurry"*, *"Limited Offer"*, and *"Only Few Left"*. These words are used to create psychological pressure and encourage users to make quick decisions.

8.3: Popup Analysis:

Pattern Type	Average Popup Count
Dark	Higher
Normal	Lower
Overall Average	2.55

Analysis:

Dark Pattern pages show a higher number of popups compared to Normal pages. Frequent popups distract users and influence their behavior, which is one of the common characteristics of Dark Patterns.

8.4: Machine Learning Results

Metric	Result
Algorithm Used	Random Forest Classifier
Input Features	Urgency Words, Popup Count, Price Difference, Button Highlight
Output	Dark / Normal Classification
Evaluation	Confusion Matrix

Analysis:

The Random Forest model successfully classified Dark and Normal patterns using important behavioral features. The Confusion Matrix demonstrated the model's capability to correctly identify most Dark Pattern instances.

8.5: Feature Importance

Feature	Importance in Prediction
Urgency Words	High
Popup Count	High
Price Difference	Medium
Button Highlight	Medium

Analysis:

Feature Importance analysis revealed that Urgency Words and Popup Count have a greater influence on identifying Dark Patterns. These features represent common manipulative techniques used in E-Commerce websites.

Overall Analysis

The project demonstrates that Dark Patterns are widely used in E-Commerce platforms and are strongly associated with urgency messages, excessive popups, price manipulation, and highlighted buttons. The integration of Python for Machine Learning, MySQL for database analysis, and Power BI for visualization provided a complete end-to-end analytical framework for detecting Dark Patterns effective.

9.CONCLUSION:

This project, **"Dark Pattern Detection in E-Commerce Platforms,"** was developed to analyze and identify manipulative user interface practices that influence user decisions during online shopping. The project successfully combined **Python, MySQL, and Power BI** to create a complete end-to-end analytical solution.

The dataset was first preprocessed and cleaned using Python to remove duplicate values and handle missing data. Exploratory analysis showed that Dark Patterns are significantly more common than Normal Patterns in the selected dataset. The analysis revealed that features such as **urgency words, popup count, price difference, and button highlighting** play an important role in identifying manipulative design practices.

Machine Learning techniques were further applied using the **Random Forest Classifier** to classify Dark and Normal Patterns. The Confusion Matrix and Feature Importance analysis helped in understanding the prediction performance and identifying the most influential features in Dark Pattern detection.

MySQL was used to store and analyze the dataset efficiently, while Power BI dashboards provided interactive visualizations and KPI summaries. The dashboards clearly demonstrated the dominance of Dark Patterns and highlighted the behavioral factors associated with them.

Overall, this project provides a practical approach for detecting unethical design strategies used in E-Commerce platforms and contributes to improving transparency and user awareness in digital environments.

FUTURE SCOPE:

• Real-Time Dark Pattern Detection

The current system works on a static dataset. In the future, the project can be extended to detect Dark Patterns in real time by analyzing live E-Commerce websites and webpages dynamically.

• Advanced Machine Learning Models

More advanced Machine Learning and Deep Learning algorithms such as **XGBoost, LightGBM, Neural Networks, and Transformer-based models** can be used to improve prediction accuracy and model performance.

• Natural Language Processing (NLP)

Future versions of this project can apply NLP techniques to analyze webpage text more effectively. This will help in automatically detecting manipulative phrases, hidden messages, and misleading promotional content.

• Browser Extension Development

The system can be converted into a browser extension that automatically warns users whenever Dark Patterns are detected while browsing E-Commerce websites. This would provide practical real-world usability.

• Large Scale Dataset Integration

The project currently uses a dataset of 1000 records. Future research can include larger datasets collected from multiple E-Commerce platforms to improve model generalization and reliability.

• Explainable AI (XAI)

Explainable AI techniques can be integrated to provide clear explanations for each prediction made by the model. This would improve transparency and help users understand why a webpage is classified as a Dark Pattern.

• User Awareness Applications

The developed model can be integrated into educational applications or consumer protection systems to increase awareness about manipulative online practices and promote ethical digital experiences.

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